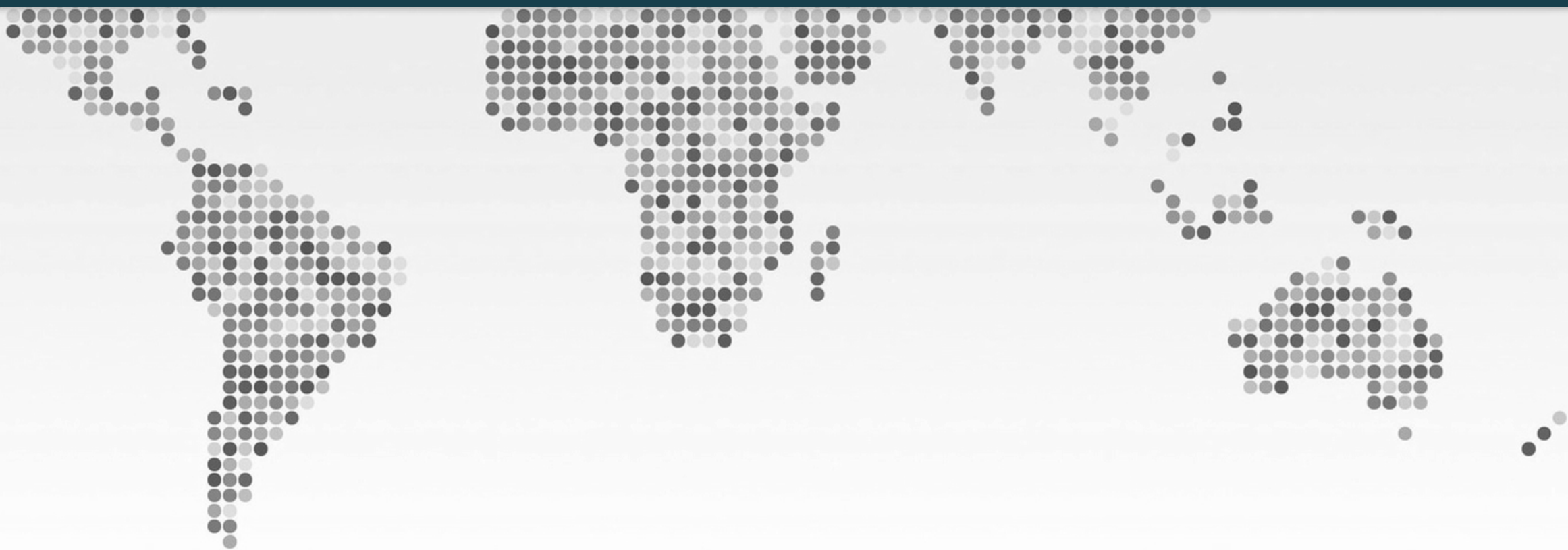


Software Engineering for Artificial Intelligence

Introduction to MLOps



Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

From the last lecture, we learned that AI-enabled systems in production need **continuous monitoring** of both system and model behavior. Operational decisions (e.g., retraining, rollback, alerting) are indeed driven by **observability data** - key challenges such as **prediction drift**, **feature drift**, and **latency degradation** must be addressed quickly to maintain performance and system usefulness.

Besides those key principles, we may want to make some additional observations:

- 1 To react to drifts and other issues that may arise in operation, we would require instruments that allow us to speed up debugging. Metrics, logs, and traces play a fundamental role in tracking and alerting developers of potential issues, but they are not sufficient unless the system also guarantees **reproducibility**.

Reproducibility. The ability to *reproduce a model, its predictions, and performance* exactly, given the same data, code, configuration, and environment.

Reproducibility would allow data scientists to debug and trace back production failures or explain unexpected predictions. It would also ensure regulatory compliance (e.g., GDPR, auditing) and service-level agreement verification.

As a side effect, reproducibility supports collaborative development, as others can rerun the work and get the same results.

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Besides those key principles, we may want to make some additional observations:

- 2 To maintain AI systems effectively, multiple roles—from data scientists and ML engineers to DevOps and product owners—must contribute across the entire lifecycle. Metrics, pipelines, and infrastructure can support their individual tasks, but they are not sufficient unless the system also enables **collaboration**.

Collaboration. The ability of *data scientists, ML engineers, DevOps engineers, and stakeholders* to work together effectively on the ML system throughout its lifecycle.

AI-enabled systems are cross-functional by nature: data scientists build models, engineers deploy them, product teams monitor outcomes.

Misalignment can lead to production failures, delays, or misinterpreted metrics. In addition, collaboration improves feedback loops, model quality, and operational agility, hence making the overall system more efficient.

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Besides those key principles, we may want to make some additional observations:

- 3 To keep AI systems robust as data volumes grow, use cases evolve, and workloads shift, we need to ensure that our solutions remain efficient and responsive. Optimization of individual model components helps, but it is not sufficient unless the system also guarantees **scalability**.

Scalability. The system's ability to handle *increasing amounts of data, users, or complexity without degradation* in performance or maintainability.

AI models often need to scale with growing datasets and inference traffic.

Teams may need to train on large distributed data or deploy models across geographically distributed infrastructures (e.g., edge, cloud, hybrid). Without scalability, costs increase and user experience suffers.

Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

MLOps = From Observation to Action

AI in Operation was about **seeing** what's going on. **MLOps** is about **acting** on it automatically, efficiently, and reproducibly.

Operational Need

- Detect and react to drift
- Ensure model quality in production
- Reproduce models for audit/debug
- Deploy with low risk
- Coordinate data scientists & engineers
- Manage experimentation at scale

MLOps

- Monitoring & automated retraining
- Continuous evaluation & V&V
- Versioning of code, data, and config
- CI/CD pipelines
- Shared infrastructure, automated flow
- Experiment tracking, model registry

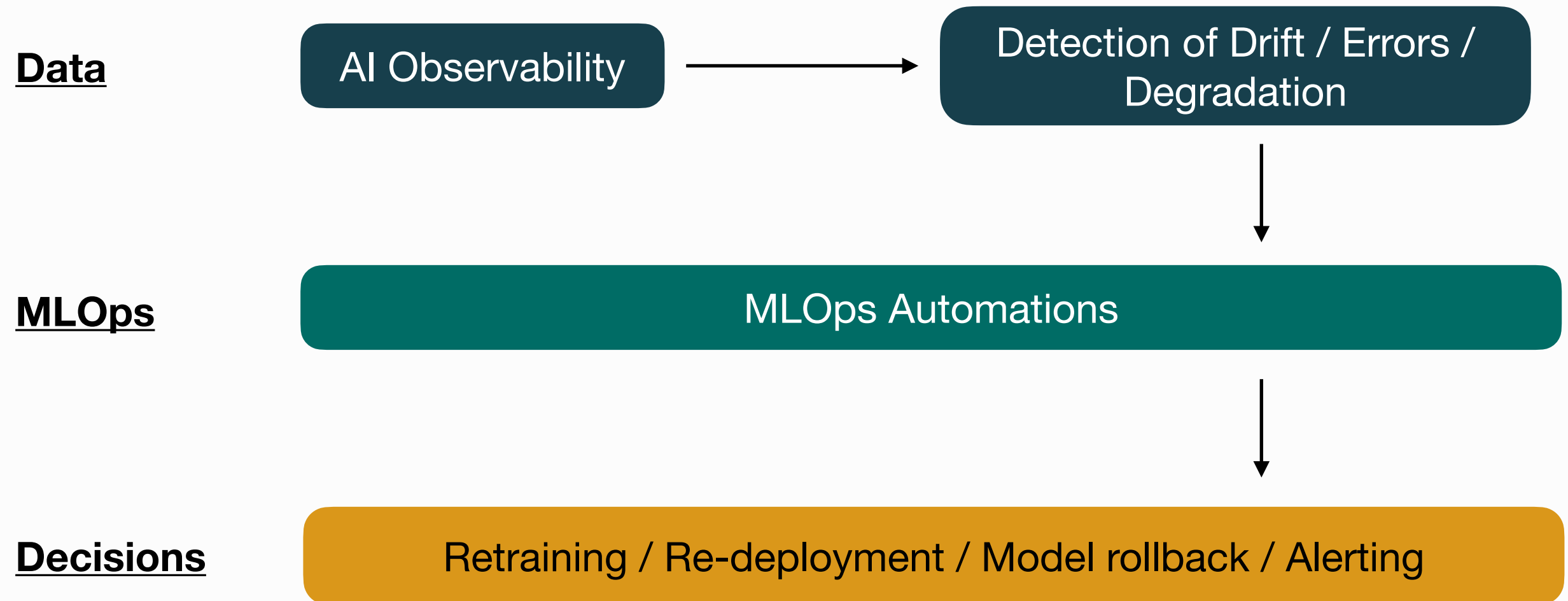
In other terms, MLOps fully implements software engineering principles to AI-enabled systems, allowing **auto-monitoring**, **structured retraining**, and **version-controlled experiments**. Reliable AI is grounded on MLOps: we can even refine the equation in: $[\text{Model Lifecycle}] + [\text{Observability}] + [\text{Automation}] + [\text{Collaboration}] = \text{MLOps}$.

Software Engineering for Artificial Intelligence: Introduction to MLOps

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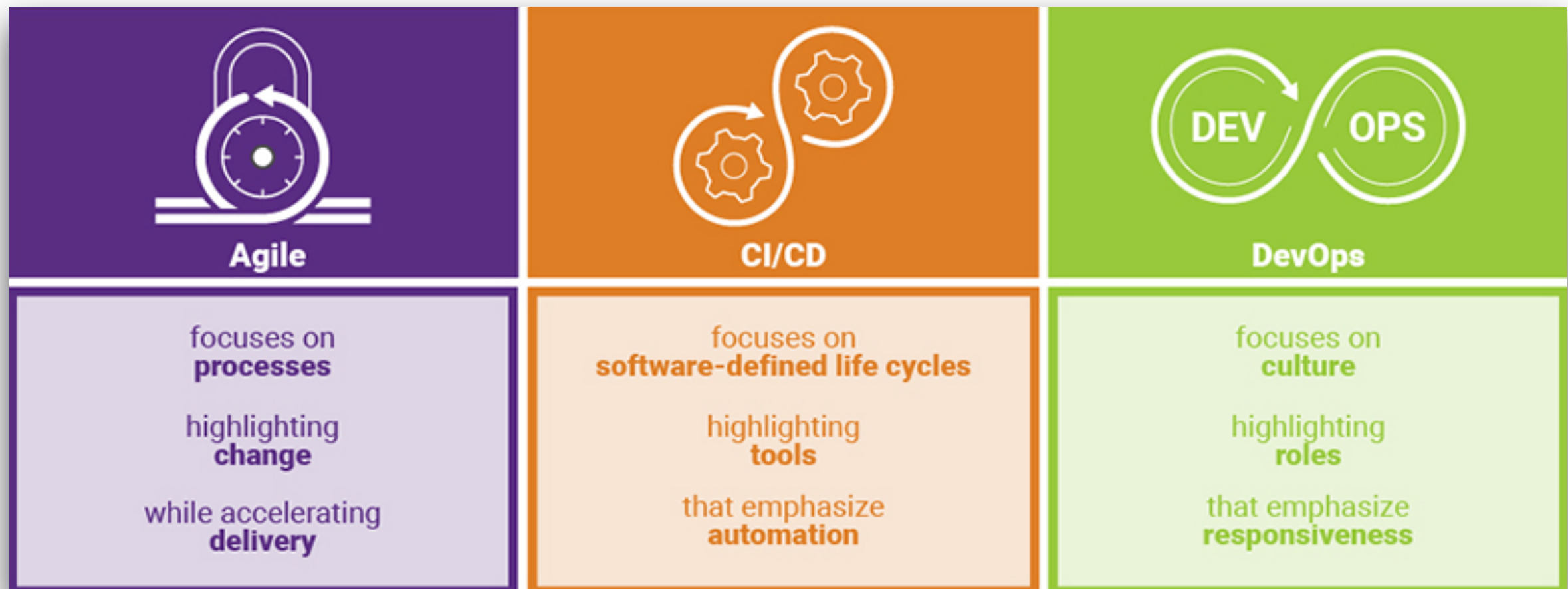
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Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

MLOps comes from three well-known software engineering principles.



Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

Agile software engineering is an iterative and collaborative approach to developing software that emphasizes adaptability, customer feedback, continuous delivery, and working solutions over rigid planning and documentation.

In this approach, work is done in short time frames called **iterations** or **sprints** (usually 1–4 weeks). After each sprint, a potentially shippable version of the product is delivered.

Stakeholders are involved throughout the process. Requirements can evolve based on feedback, not locked down upfront. Plans are flexible. If something does not work, the team can change direction quickly.

Developers, testers, designers, and sometimes users work together. Teams are self-organizing and **empowered by tools** to make decisions.

While this is an organization-, process-level transformation of the work activities, the core principle is **automation**: we want to govern continuous changes by making ourselves agile, but **true agility depends on continuous feedback** to adapt effectively.

So, agility focuses on improving development processes by **embracing** change and **accelerating** delivery to meet evolving needs. This is key in AI-enabled systems because of their dynamic nature—models degrade over time, data evolves, and new requirements frequently emerge, demanding fast iteration, frequent updates, and continuous validation.

Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

Let's do an example.

Product owner introduces a user story: *“As a user, I want to see personalized product recommendations based on my browsing and purchase history”*

The team breaks this down into tasks:

1. Update data pipeline to include recent user activity.
2. Train a new collaborative filtering model.
3. Integrate model into backend service.
4. Display top 5 recommendations on homepage.

Two-week sprint begins:

- The **data scientist** retrains the model on recent data.
- The **backend engineer** wraps the model in a REST API.
- The **frontend developer** updates the homepage UI.
- The **QA engineer** defines tests for accuracy and performance.
- **Daily stand-ups** help coordinate progress and uncover blockers.

Mid-sprint adjustment:

- Midway through the sprint, the team notices the model's accuracy drops for new users (**cold start** problem).
- They add a quick task to include trending products as fallback recommendations they adapt the backlog in real time.

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Sprint review:

- The new model is demoed in staging: it shows 20% improved click-through rate when testing it in production.
- Product owner is satisfied and approves deployment.

Sprint retrospective:

- Team discusses what worked well (tight collaboration), what did not (late detection of cold start issue), and how to improve (include a checklist for cold-start mitigation in future models).

A similar process naturally emphasizes **continuous communication** among different roles, who are encouraged to meet daily to share progress, surface blockers, and align on goals —fostering transparency, coordination, and shared ownership across the team.

In addition, it actively promotes **collaboration** by creating structured opportunities where team members must engage, align priorities, and make collective decisions.

Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

Continuous integration and delivery. Agile is about delivering **working software frequently**, responding to feedback, and adapting fast. CI/CD enables frequent releases, ensures early quality through automated tests, accelerates feedback loops, fosters collaboration, and avoids last-minute integration issues.

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CI

- The **data scientist** pushes the new model and training script to the shared repository.
- The **backend engineer** updates the model-serving API to support new inputs.
- The **CI pipeline** is triggered and any issues are flagged and shown in the CI dashboard.

CD

- Once all tests pass, the CD pipeline **builds** the Docker container with the new model API and deploys to **staging** for final validation.
- After approval, the new model is **automatically deployed** to production.

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So, continuous integration and delivery support agility but focus on tools to emphasize the automation of activities. This is called a **software-defined lifecycle**: a development and deployment process where all stages — from coding and testing to deployment and monitoring — are codified, versioned, and automated through infrastructure and pipelines, ensuring consistency, traceability, and rapid delivery.

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Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

DevOps is a set of practices that combines software development and IT operations to enable faster, more reliable, and continuous delivery of software through automation, collaboration, and shared responsibility.

DevOps is about **culture**: developers think about production, operators participate in planning. Everyone cares about uptime, performance, and user experience.

With agility, we improve how we develop and adapt software through iteration and collaboration. With CI/CD, we automate integration, testing, and delivery to ensure fast and safe releases. With DevOps, we bring it all together—aligning development and operations to deliver, monitor, and evolve software continuously and responsively.

As such, DevOps is neither a tool nor a process: *it is a cultural and organizational mindset* that establishes a common standard for how teams **develop, deliver, and operate** software collaboratively, with a strong focus on **automation, shared responsibility, and rapid feedback**.

DevOps basically promotes the practices we have been discussing so far: **infrastructure-as-code** to provision and manage environments consistently and reproducibly; **continuous monitoring and alerting** to detect issues in real time; and **continuous, automated reactions** to changes or incidents ensuring fast recovery and adaptive evolution of the system.

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Shared Ownership

CI

CD

Continuous integration and delivery

More tools to improve responsiveness: essentially, a software-defined lifecycle extended to operation

Infrastructure code

Monitoring and Alerting

Incident Management
and Feedback Loops

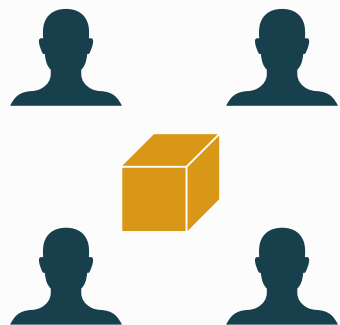
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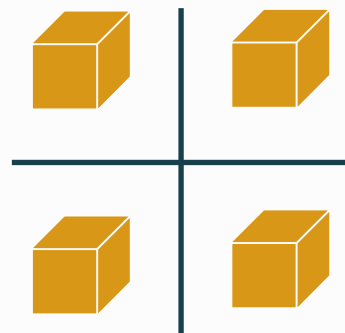
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MLOps is, again, about culture extended to models and AI algorithms. It is essentially about **agreeing to machine learning the right way and then supporting it.**

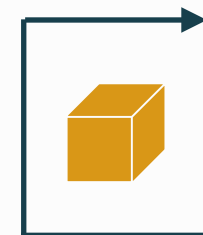
A few shared principles



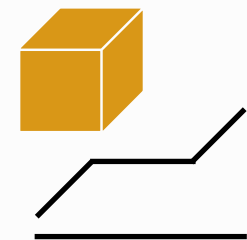
ML should be collaborative



ML should be reproducible



ML should be continuous



ML should be tested and monitored

Process

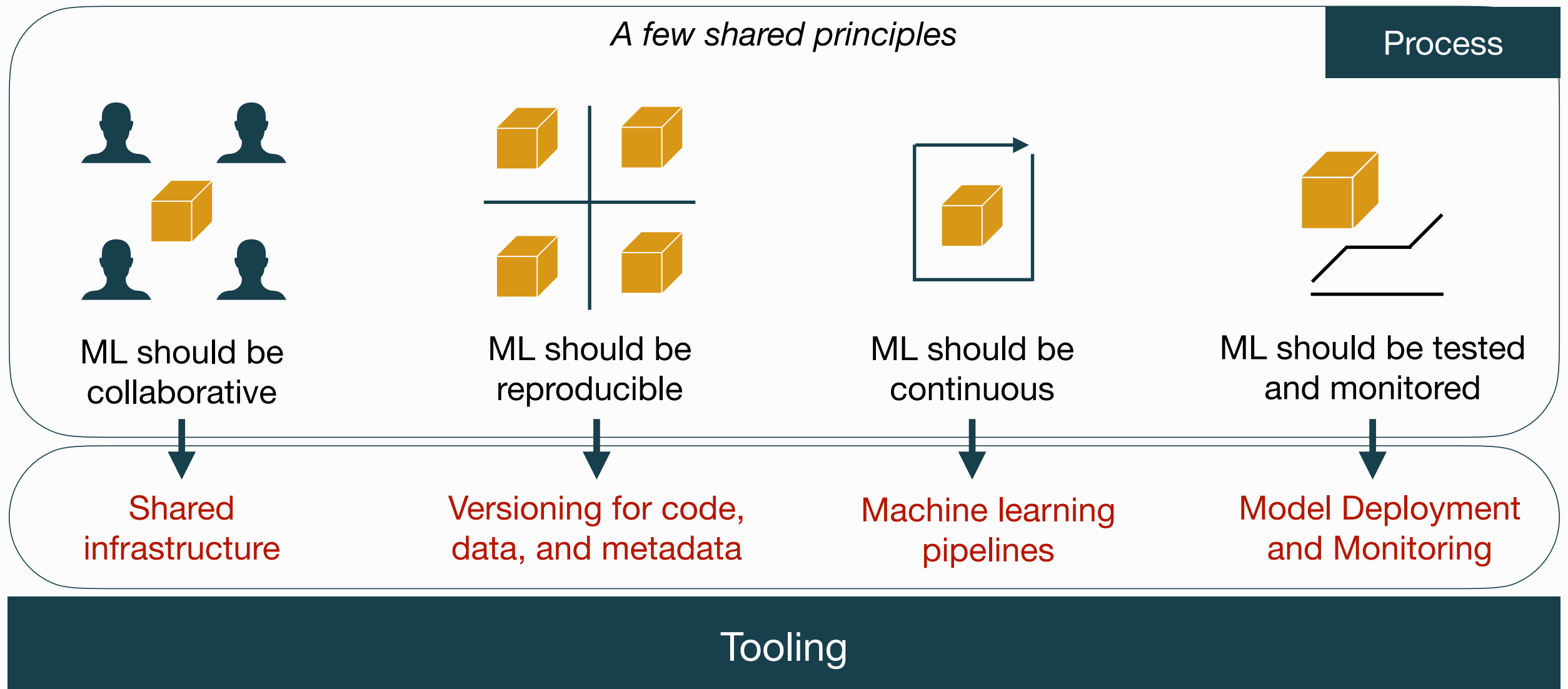
Tooling

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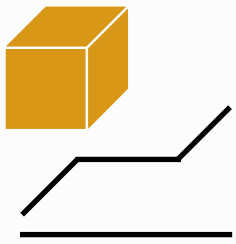
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Process



ML should be tested
and monitored



Model Deployment
and Monitoring

Socio-technical congruence. The fit between an organization's coordination requirements and an organization's social interactions.

Mentioned multiple times already, the alignment between processes and tooling lets emerge the highest and purest form of coordination, which is indeed the socio-technical congruence!

It has been shown multiple times already that a high socio-technical congruence speeds up product's delivery and significantly increases the chance of successfully satisfy the business requirements.

If this is so important, why is not MLOps largely applied? And why is not MLOps taught in a Master degree of Data Science and ML?

The answer is fairly easy: socio-technical congruence requires time, effort, software engineering skills, and management maturity!

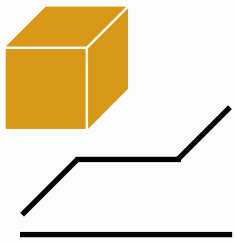
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ML should be tested
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Model Deployment
and Monitoring

Perhaps more importantly, we need to distinguish between developing a product once and being an organization aiming at making the market —> it is like a web designer versus a software engineer.

Data scientists make models: their perspective is not that of delivering a product. Hence, why shall they aim for socio-technical congruence?

Most companies cannot be defined as organization: an organization is a company with well-defined management models and that look for long-term benefits.

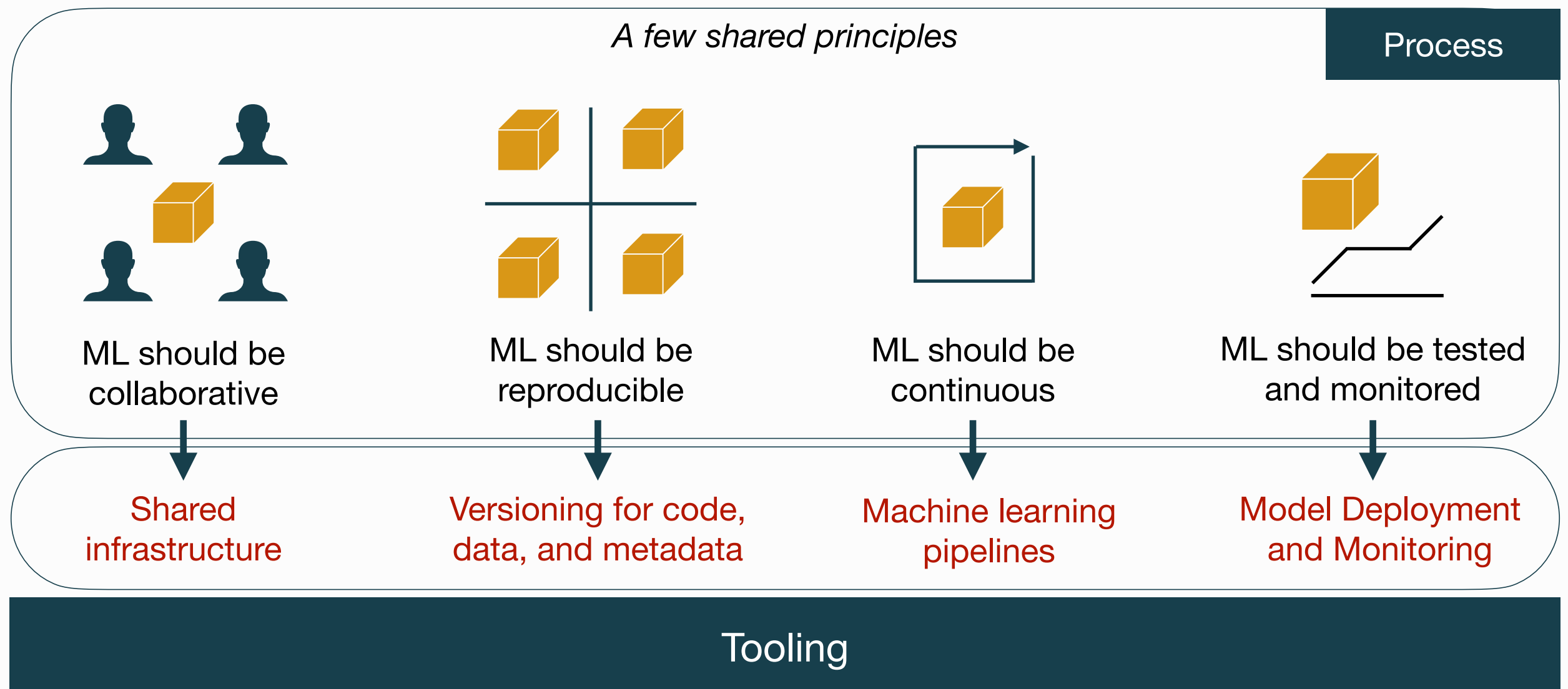
Putting in place an MLOps model requires going beyond the low-hanging fruits, sacrificing time at the beginning in favor of long-term benefits —> it is always the same trade-off: Do we want to deliver a quite-right code in the short term or do we want to deliver a product that will be easier to monitor, maintain, and evolve spending some more time and resources?

Software Engineering for Artificial Intelligence: Introduction to MLOps

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But why is putting MLOps in practice so difficult?

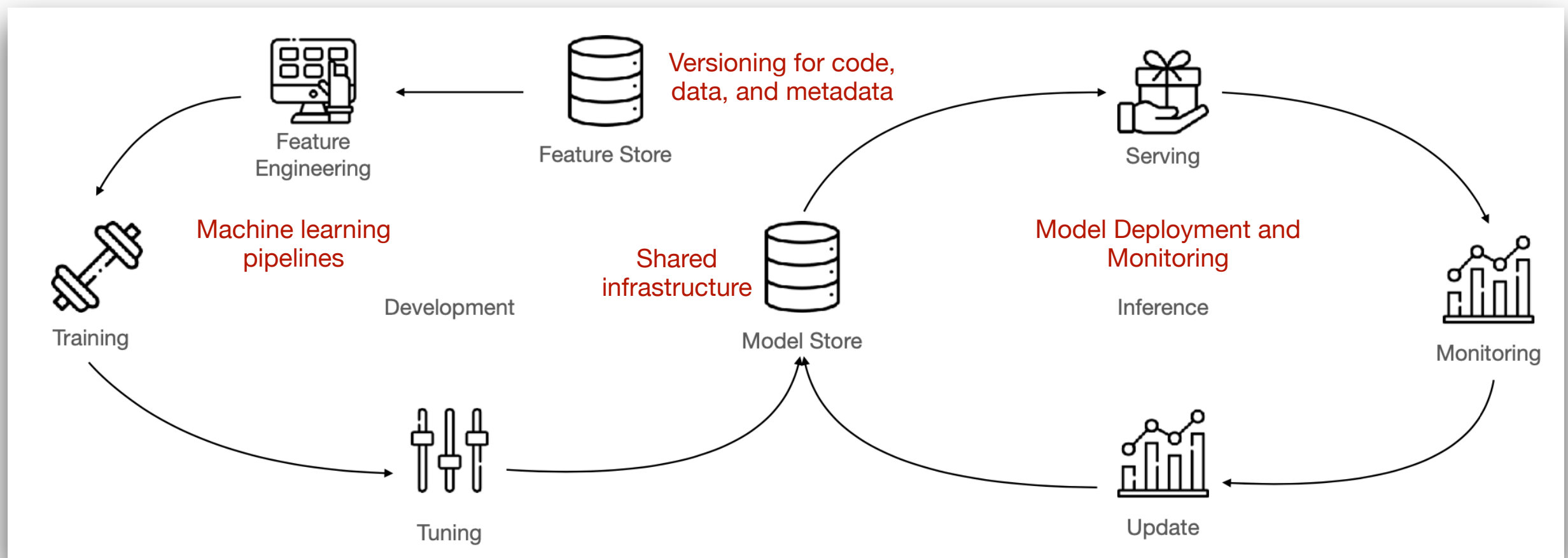


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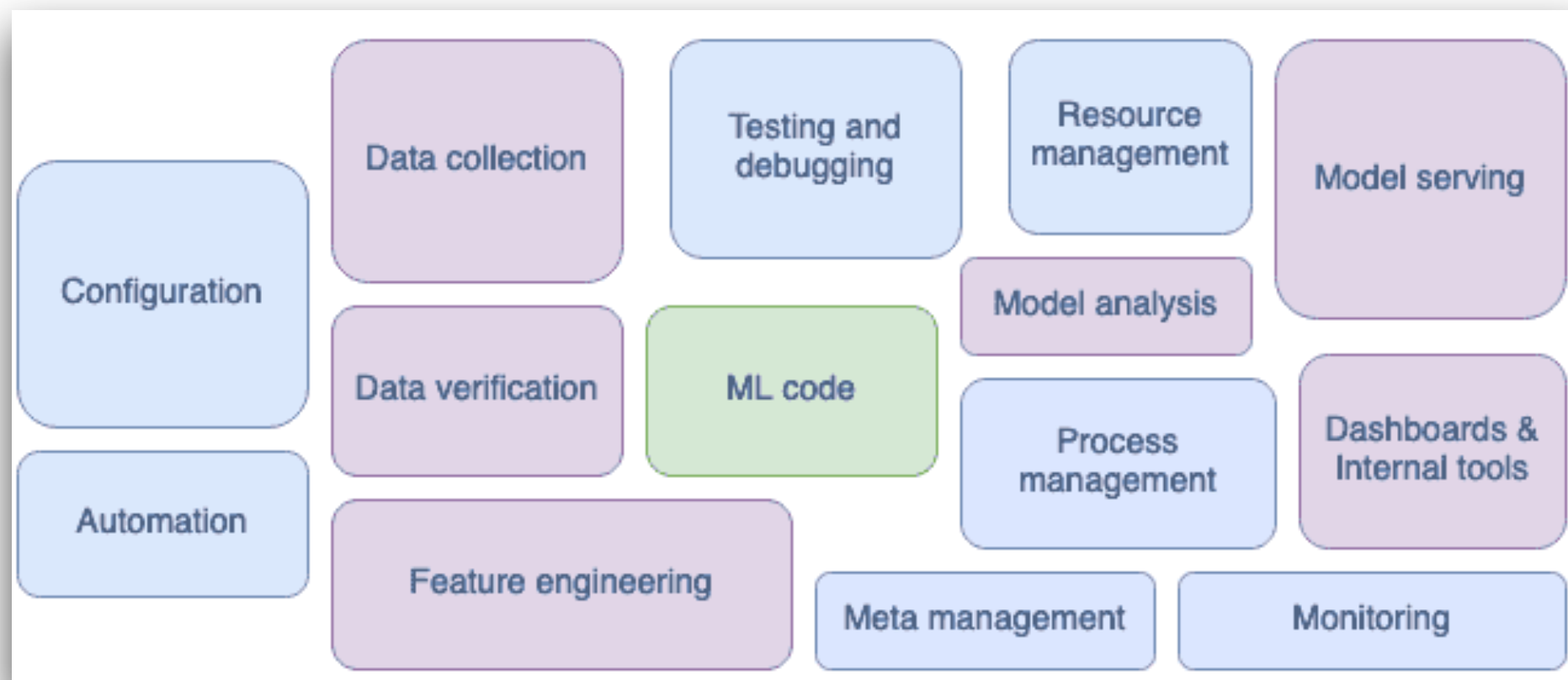
First, we need to operationalize the MLOps principles through a fully automated platform that enables a one-click model construction, deployment, and monitoring pipeline. Some have problems with setting a GitHub project... just saying...

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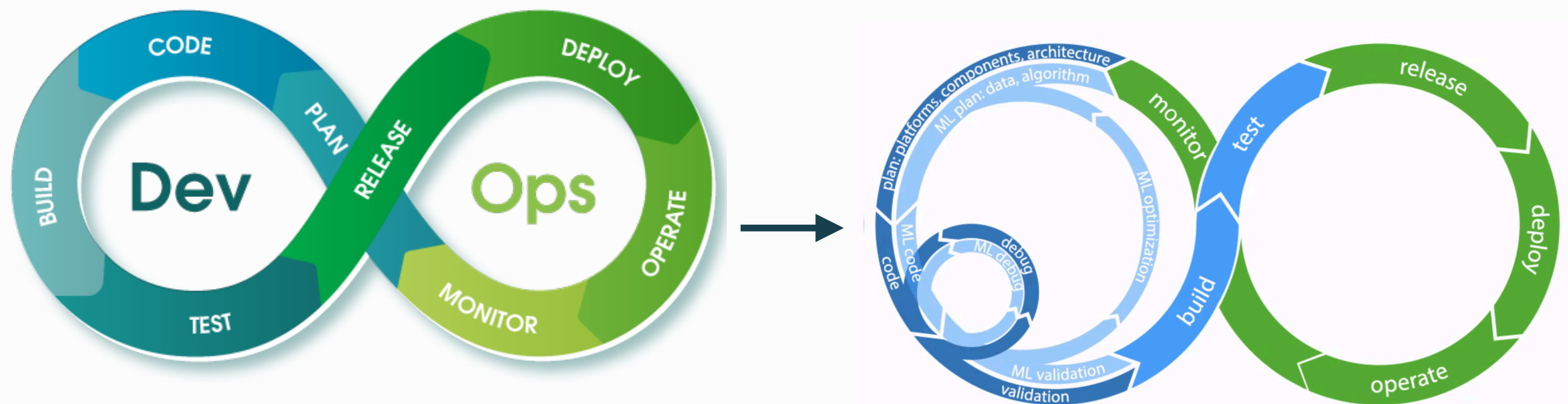
Second, a model does not live in a vacuum, but within an operating system! That is why we represent MLOps like in the figure —> MLOps is an acronym coming from *ML + Dev + Ops*, just to make clear that ML is a part of a more complex system.

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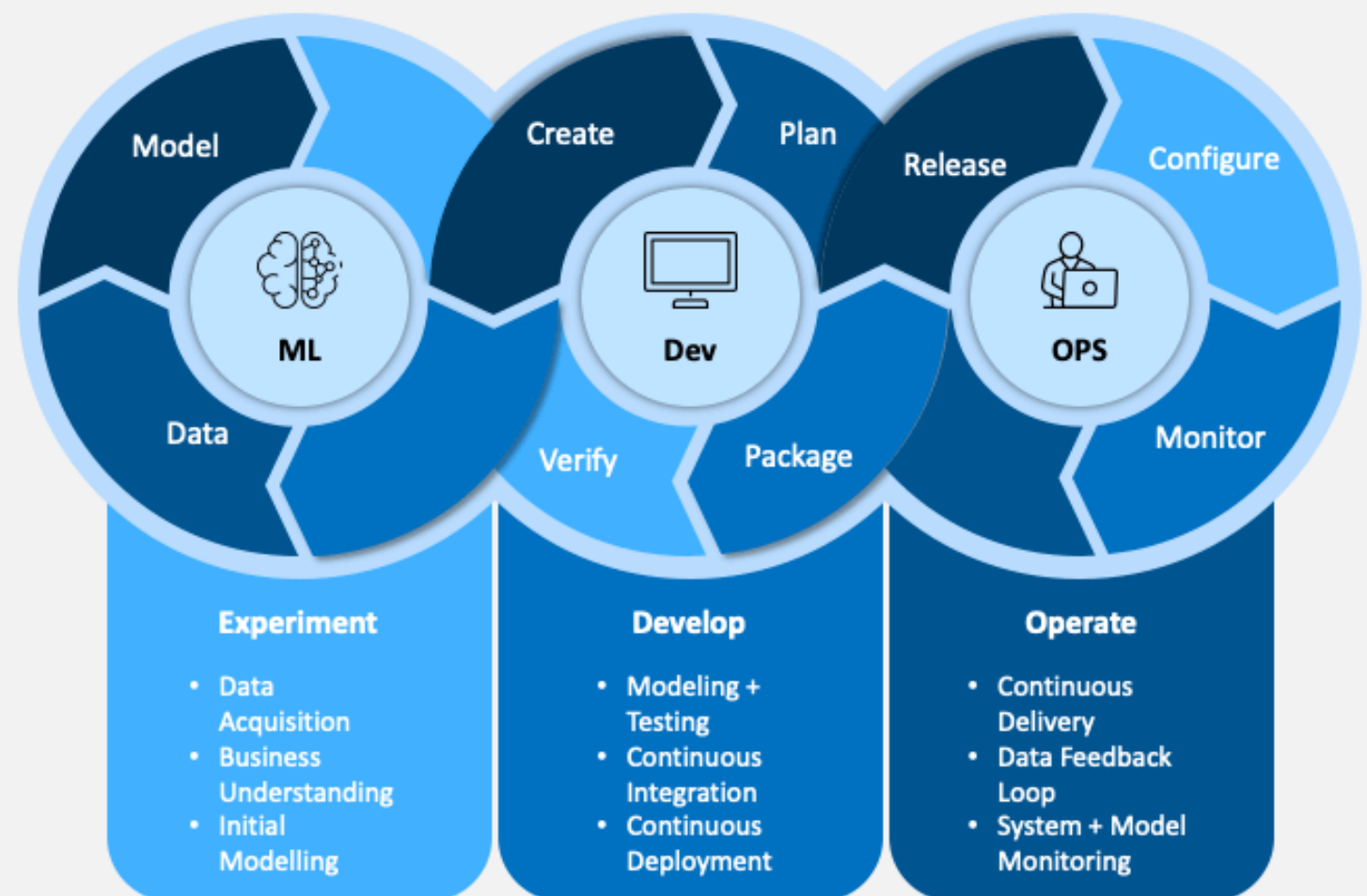
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As shown, the model is really an integral part of the whole DevOps philosophy and must communicate with other typical software engineering processes!

MLOPS

MLOPs = ML + DEV + OPS



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But why is putting MLOps in practice so difficult?

SKILLS AND EFFORT REQUIRED TO SET UP A PIPELINE CONFIGURATION

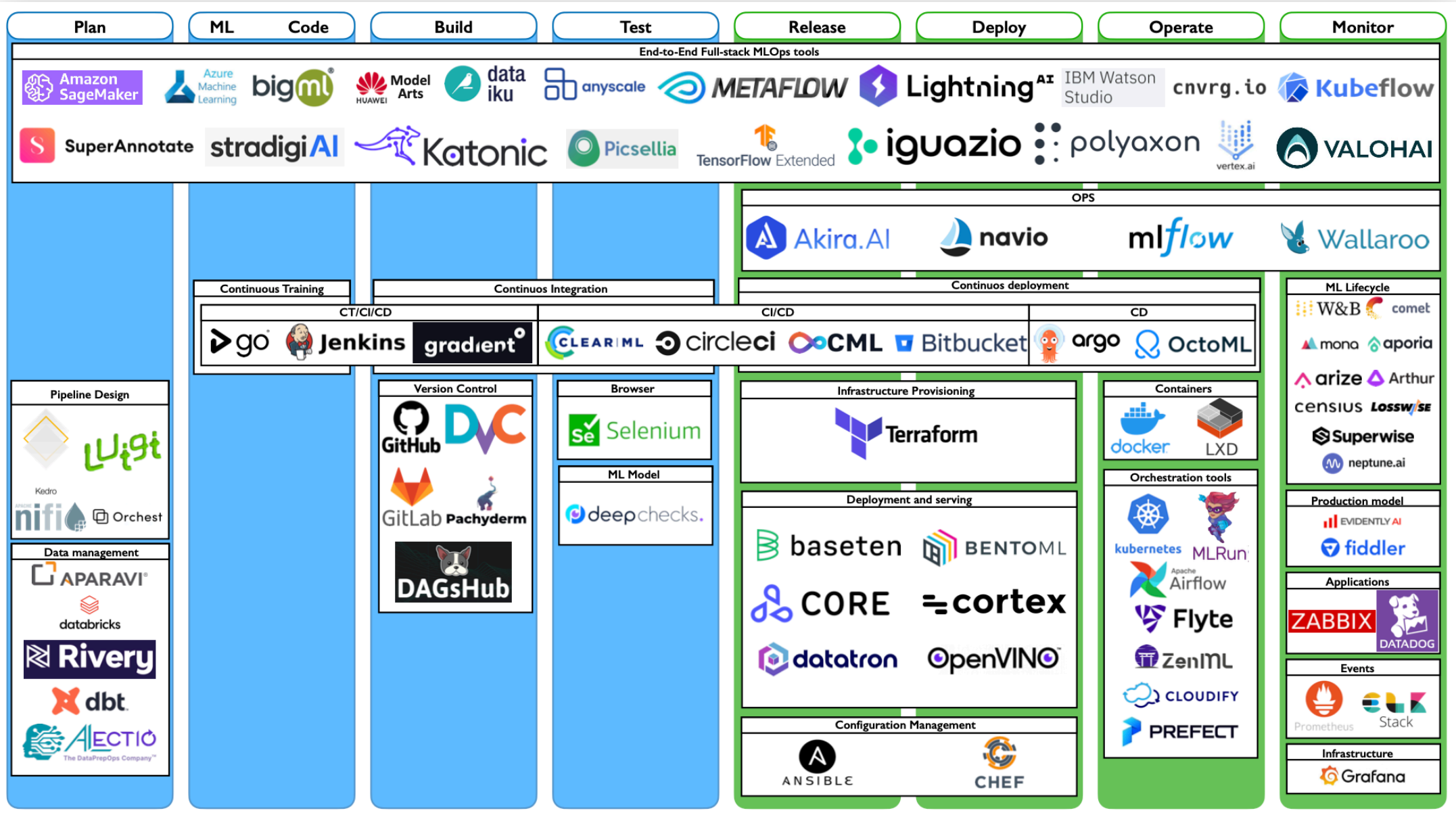
Last but not least, the biggest problem: the fully automated MLOps pipeline needs to be configured. But to reach that goal, you (1) need plenty of skills that go from machine learning till automated deployment systems and containerization and (2) need to select the right tools among the huge amount available on the market!

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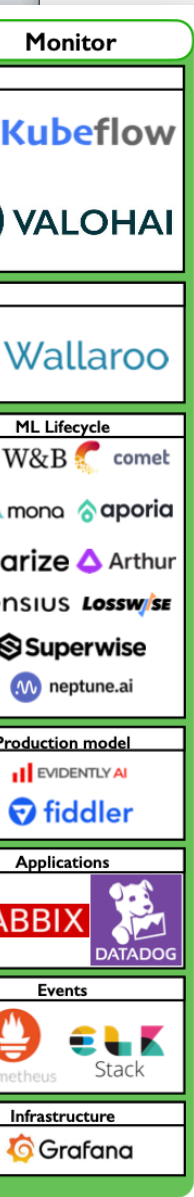
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The configuration and selection of the tools may depend on multiple factors such as: (1) the use case to fulfill, e.g., self-driving cars versus recommendation systems; (2) team and skills available; (3) regulation; and (4) other resources, e.g., bootstrapping versus well-funded.

Different roles with different skills have to interact. Do not underestimate this challenge: software engineers, data scientists, and CI/CD engineers speak different languages!

As we have been discussing, the development process is rather non-linear and very iterative, meaning that all tools and people involved might very frequently be involved on the same tasks.

Additionally, do not forget that the tools within a pipeline need to interact between each other: this may cause several technical issues, including problems with data formatting, input/output interpretation, etc.



Software Engineering for Artificial Intelligence: Introduction to MLOps

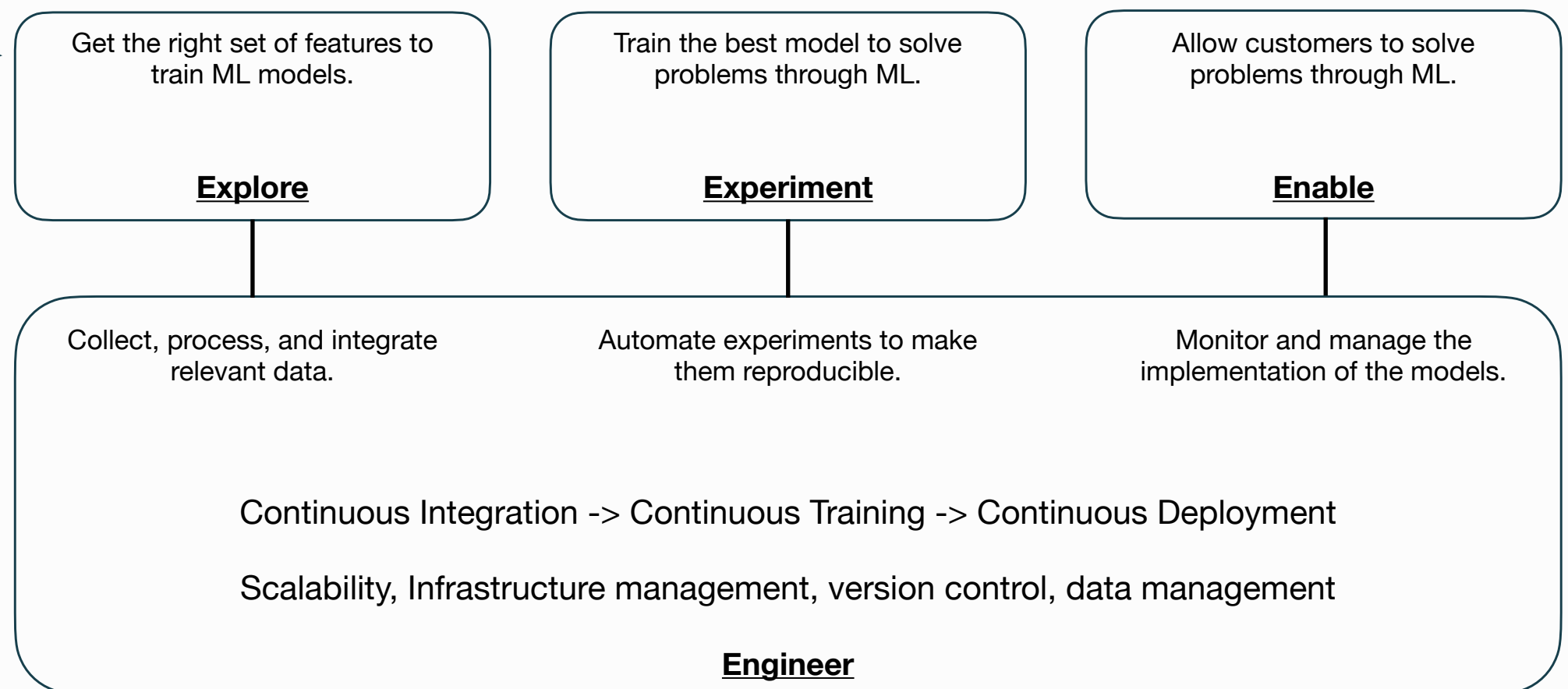
MLOps: DevOps Principles Applied to AI-Enabled Systems

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And so, if this is so complicated to put in place, why should we do that?

Automation enables the four “*E*”s of machine learning engineering:

In other terms, the availability of the MLOps infrastructure enables the engineering of the data science process.



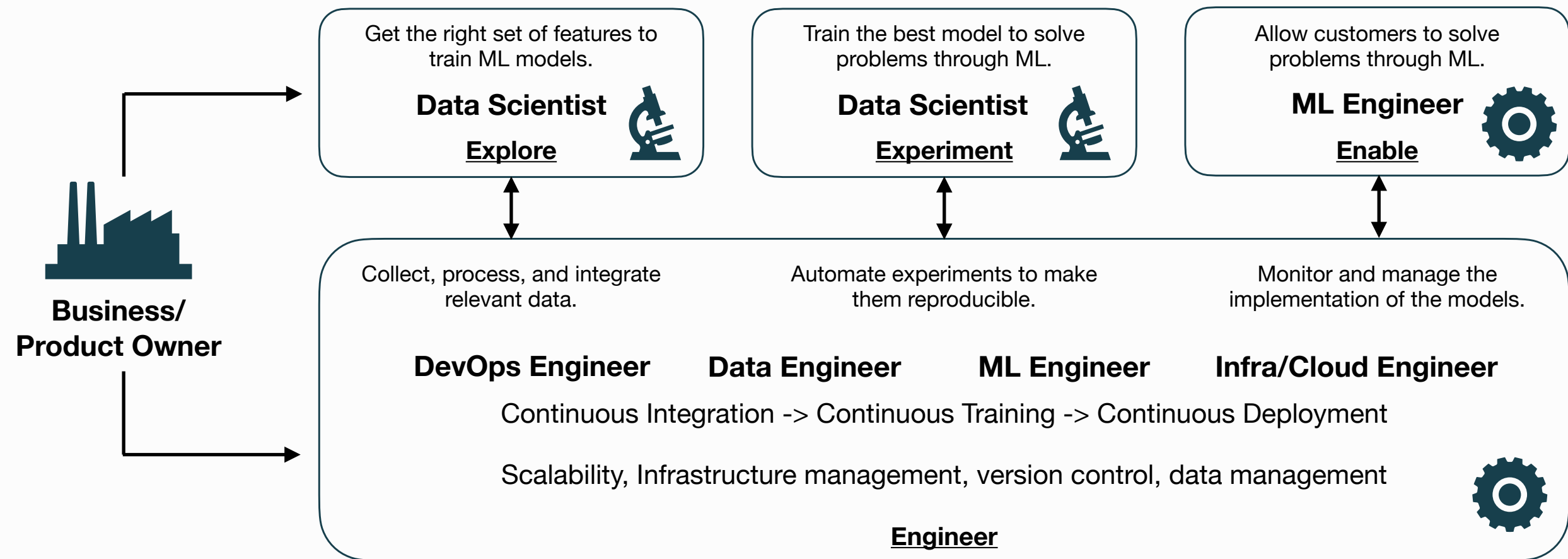
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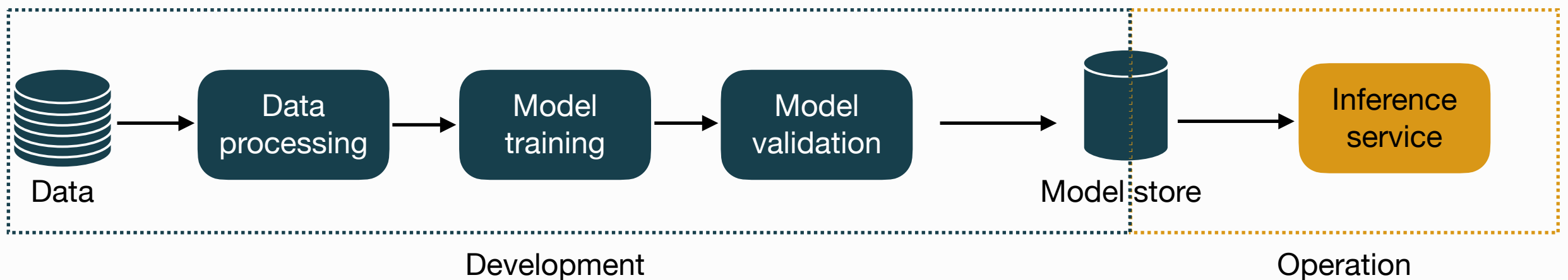


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Without MLOps



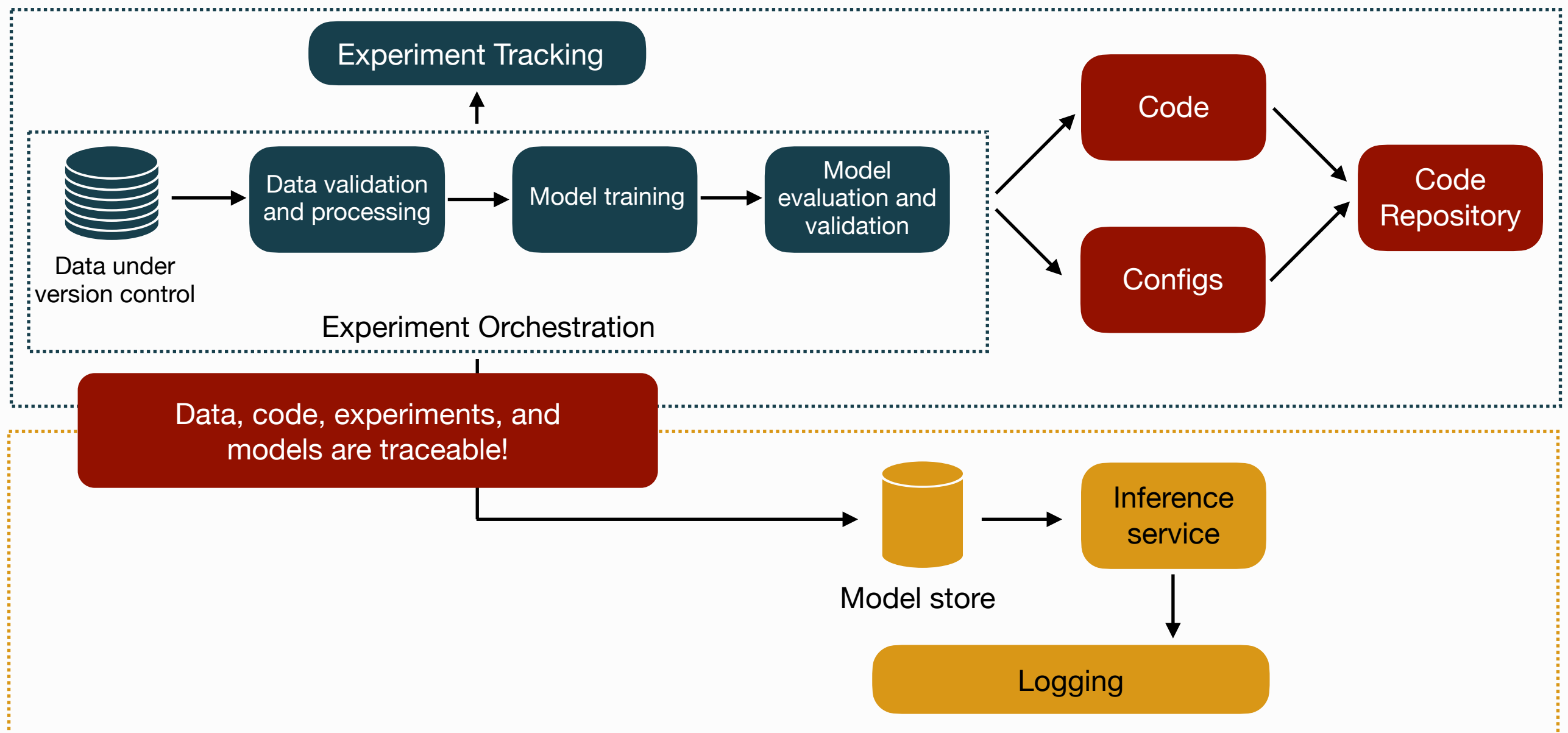
- Models are created and put into production manually.
- There is the increasing trend of using manual or notebook scripts.
- Data scientists perform experiments to optimize the model within their development environment manually.

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With MLOps - Maturity Level 0

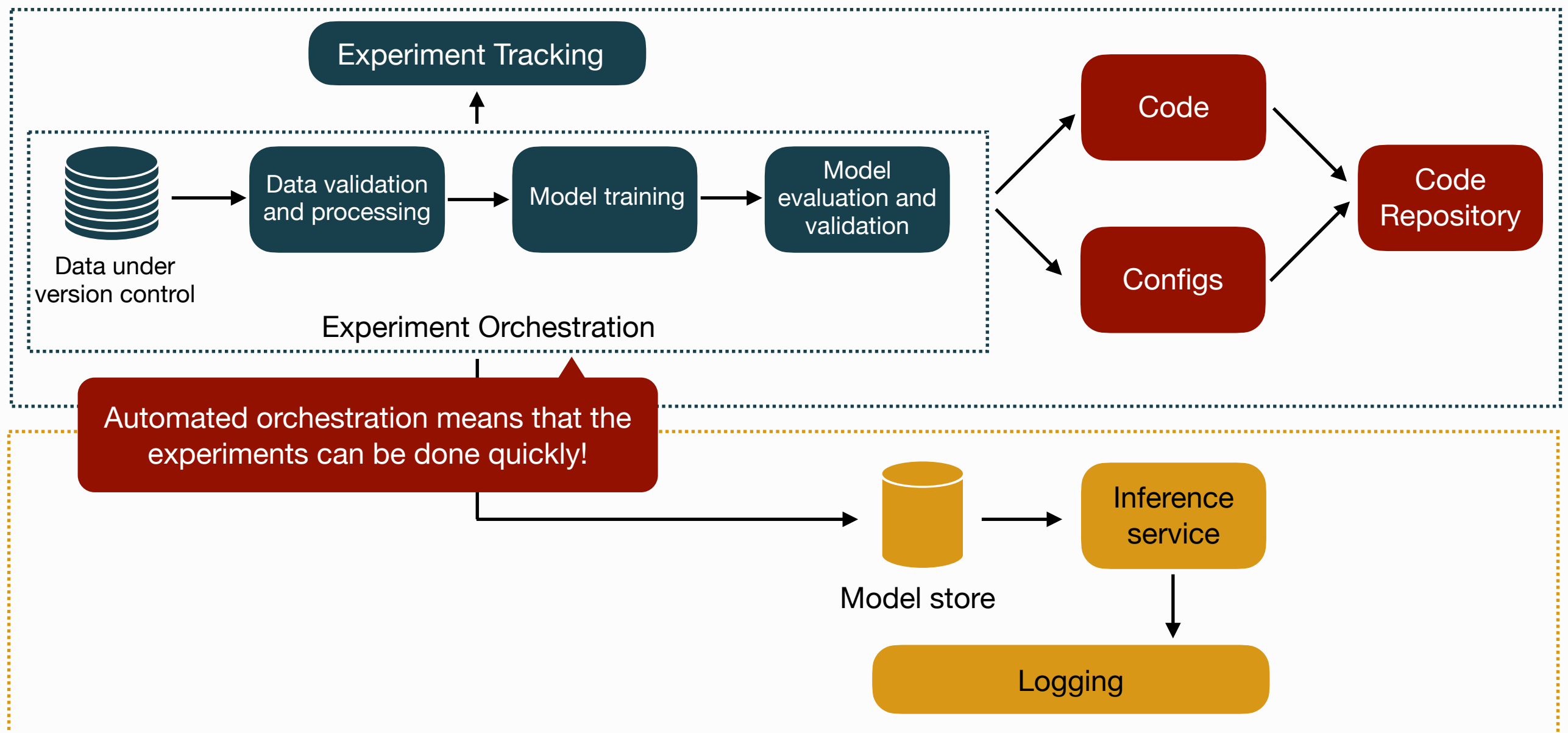


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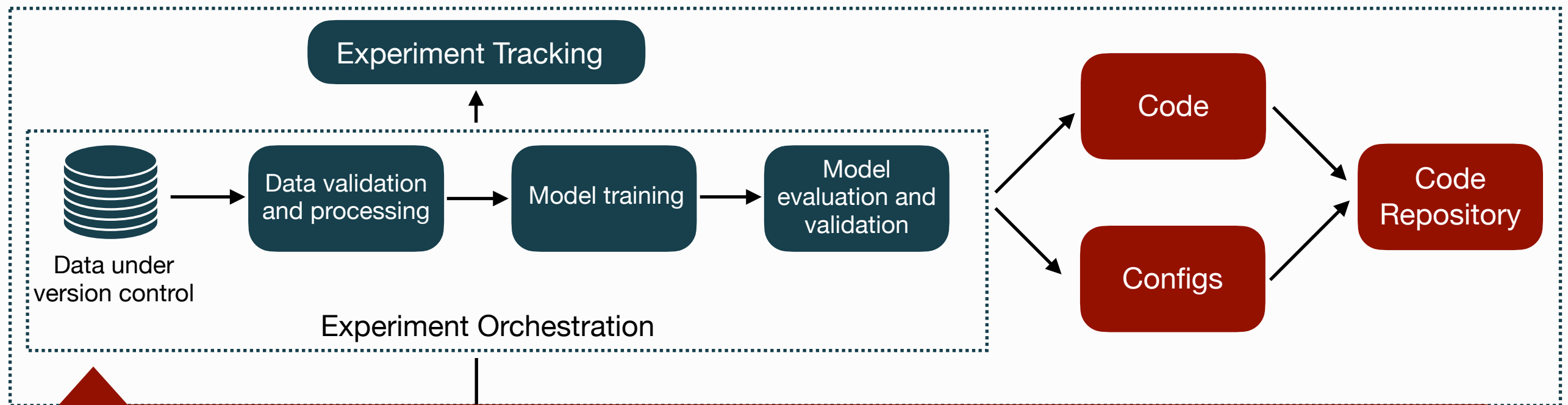


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With MLOps - Maturity Level 0



AutoML (Automated Machine Learning) refers to the process of **automating the end-to-end tasks** of applying machine learning to real-world problems — from data preprocessing and feature engineering to model selection, training, hyperparameter tuning, and evaluation.

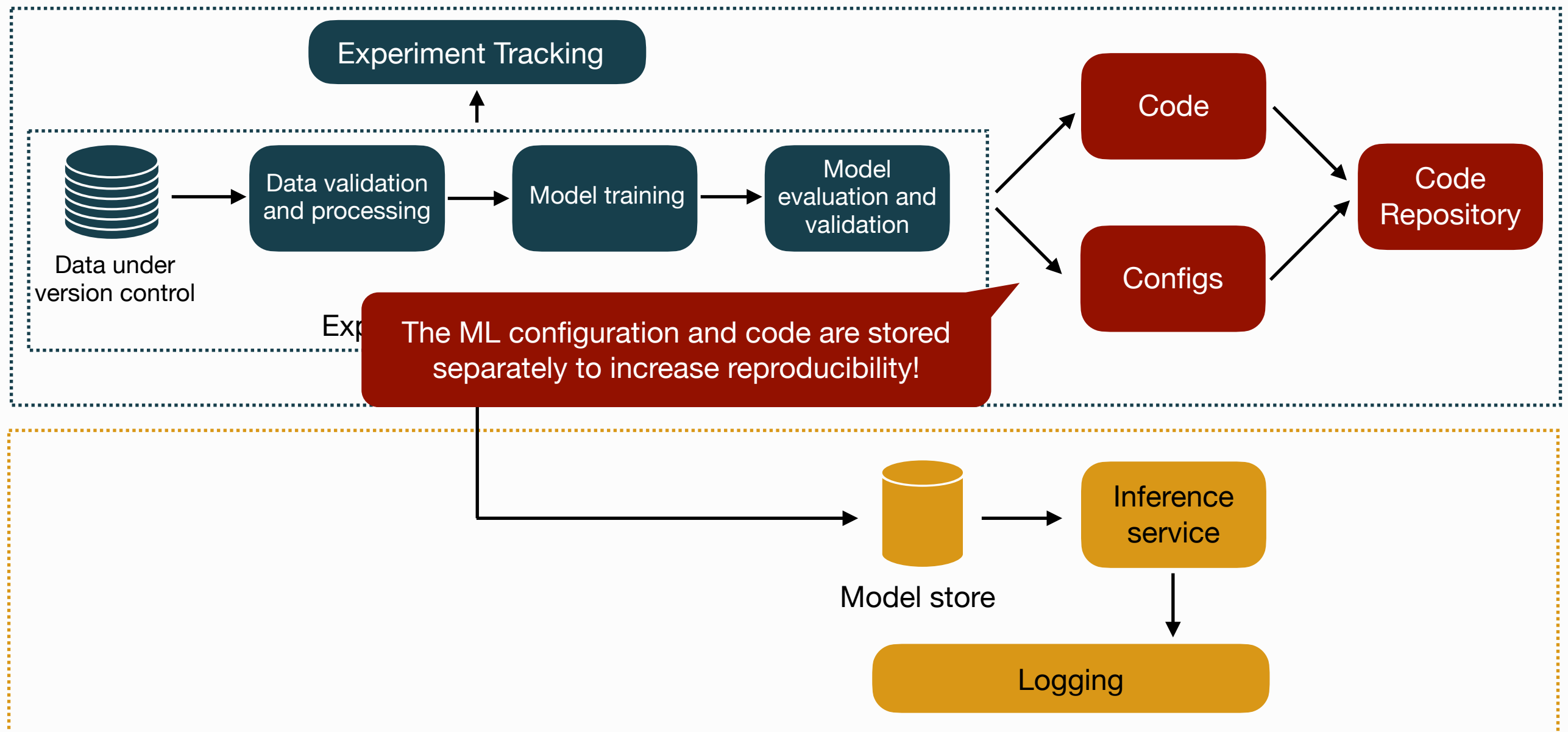
AutoML is closely tied to MLOps: by promoting continuous experimentation for reproducibility, MLOps provides the infrastructure and processes needed to **track, validate, and operationalize the models produced by AutoML.**

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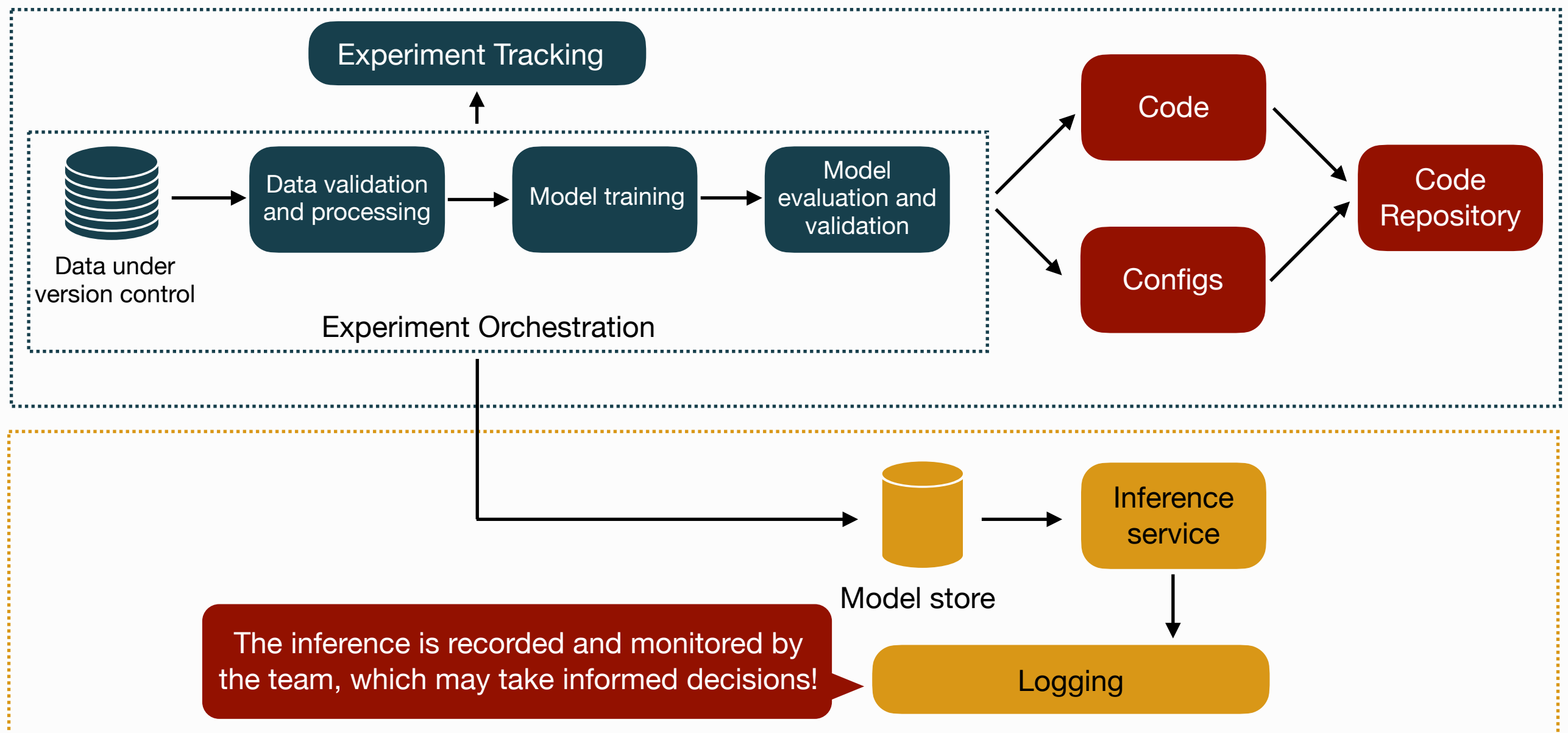


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With MLOps - Maturity Level 0

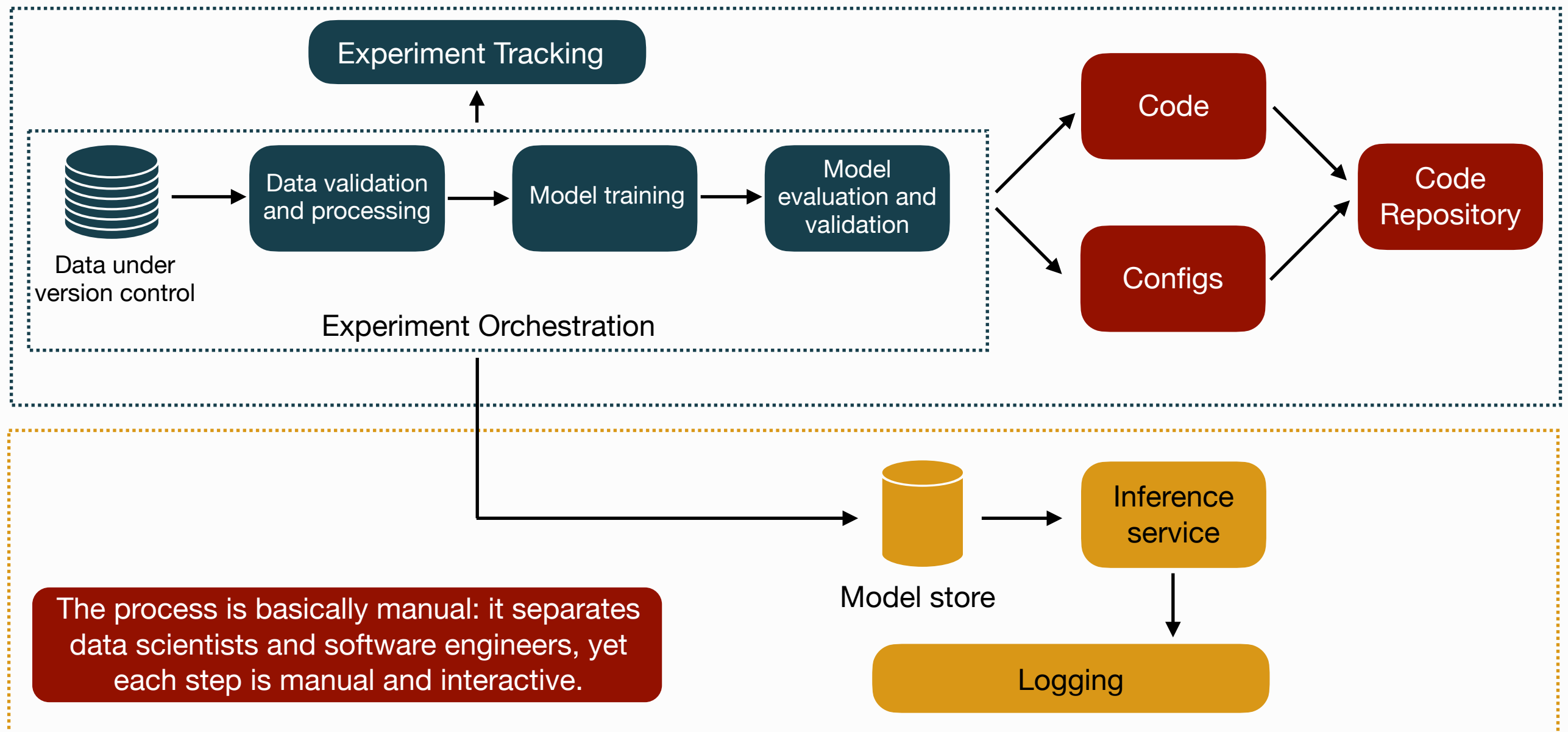


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With MLOps - Maturity Level 0

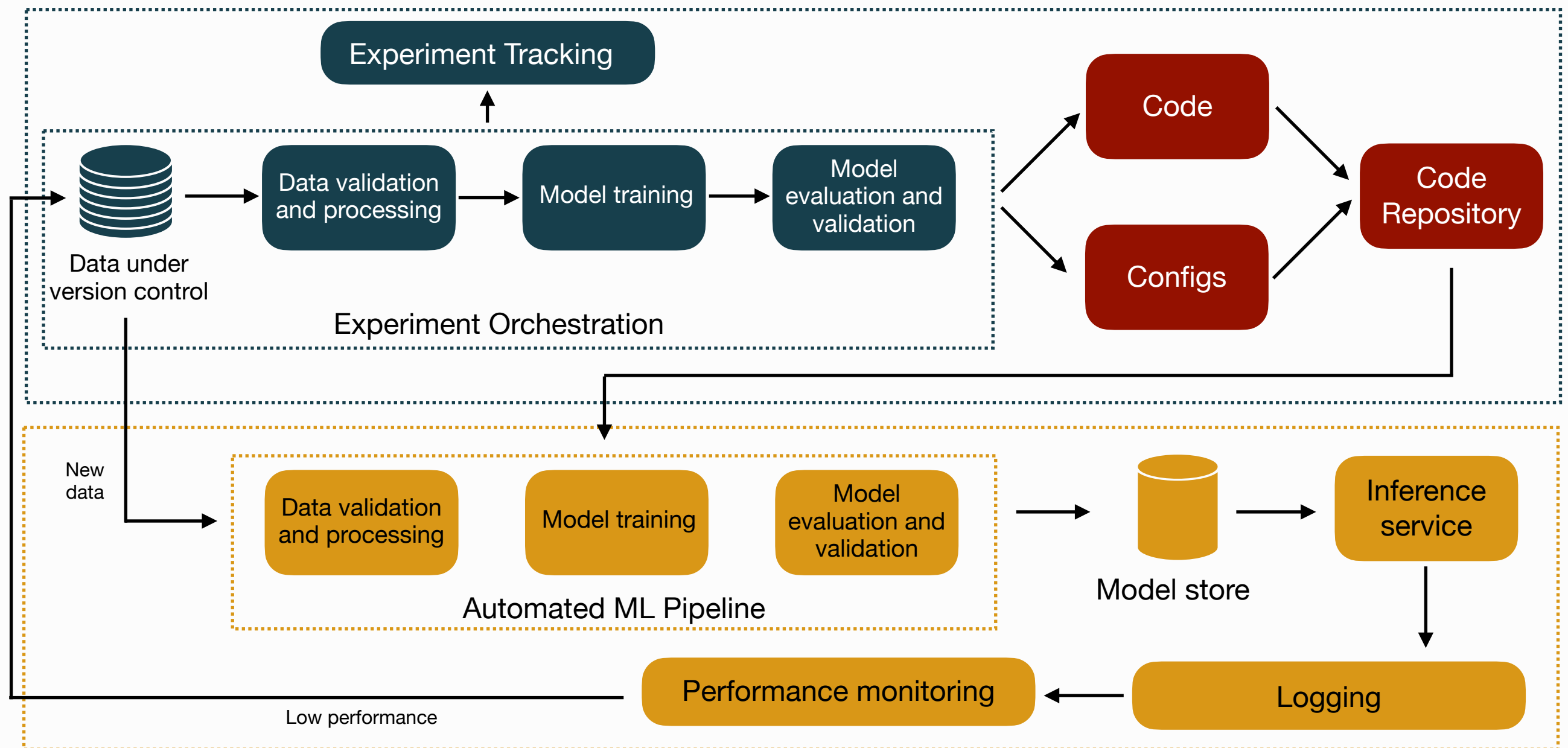


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With MLOps - Maturity Level 1

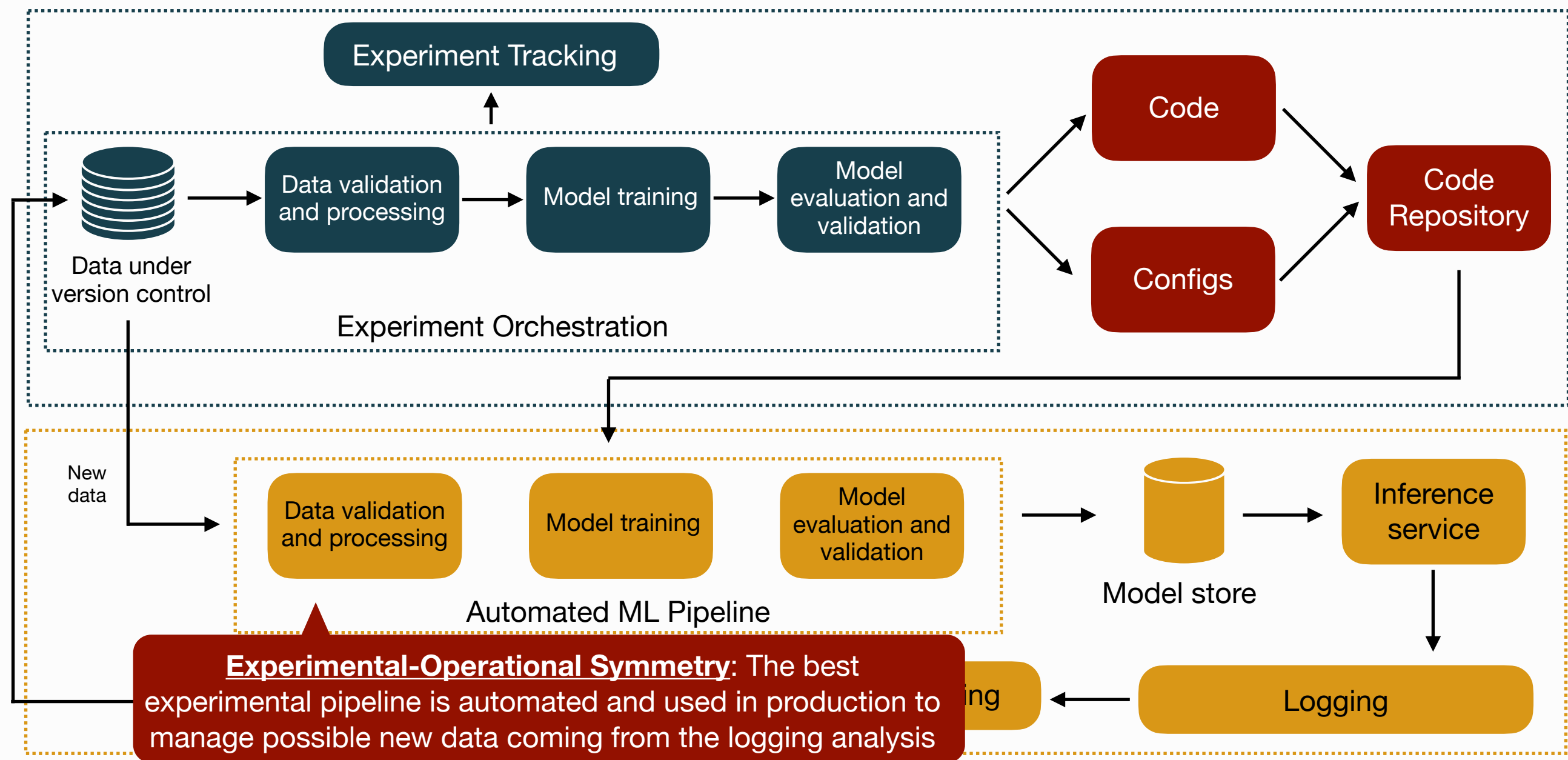


Software Engineering for Artificial Intelligence: Introduction to MLOps

MLOps: DevOps Principles Applied to AI-Enabled Systems

MLOps. MLOps refers to a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

With MLOps - Maturity Level 1

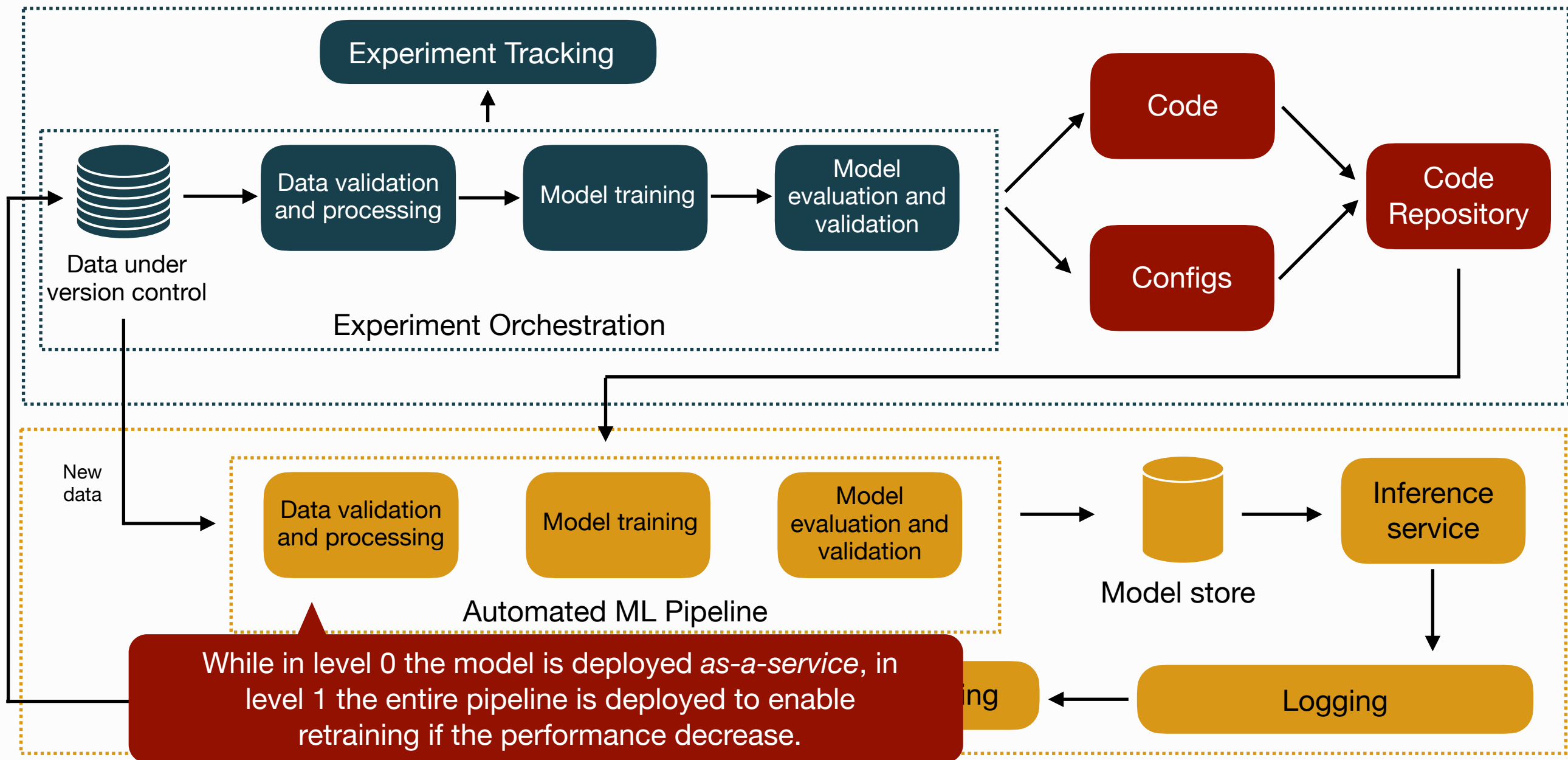


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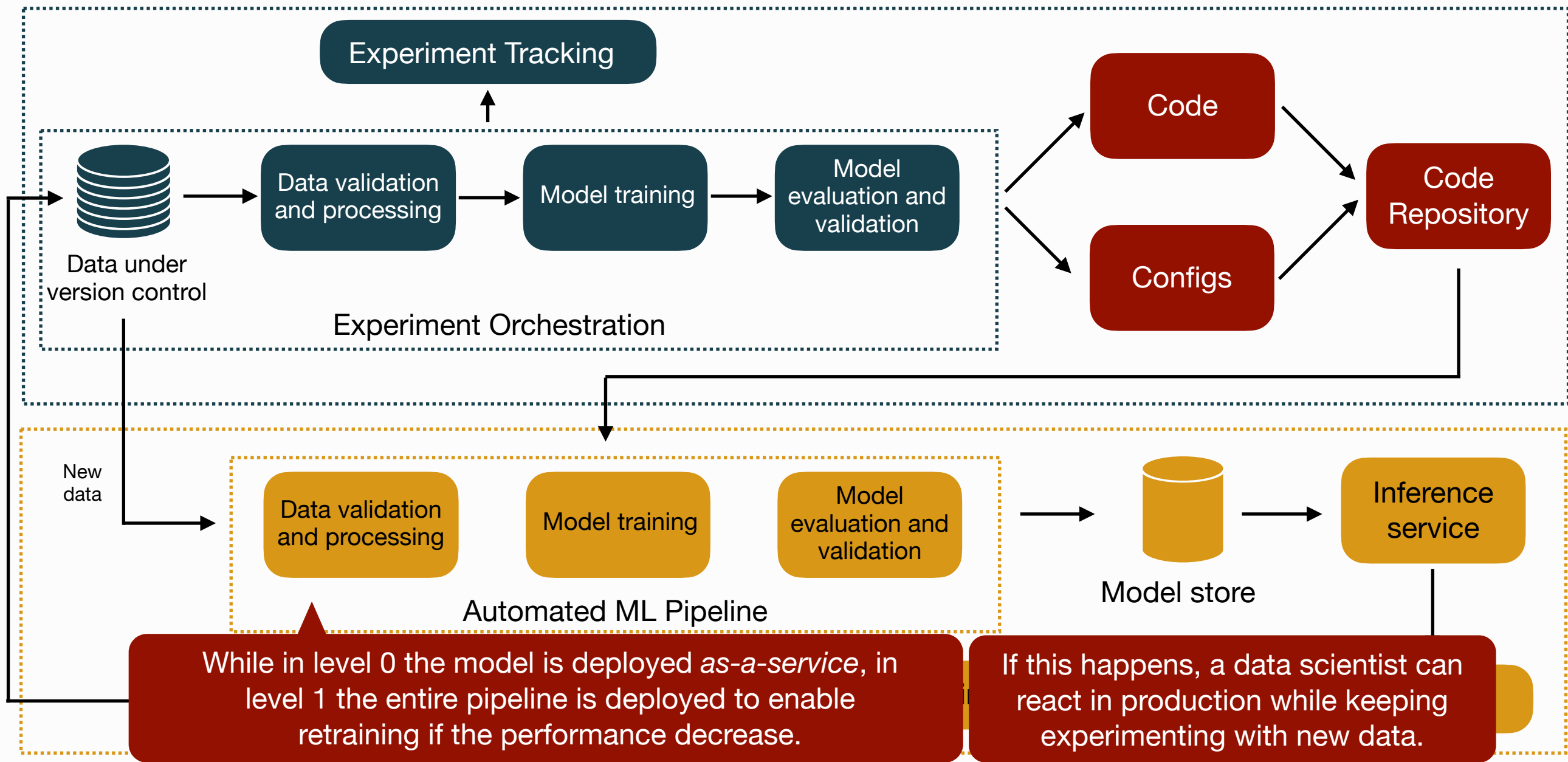


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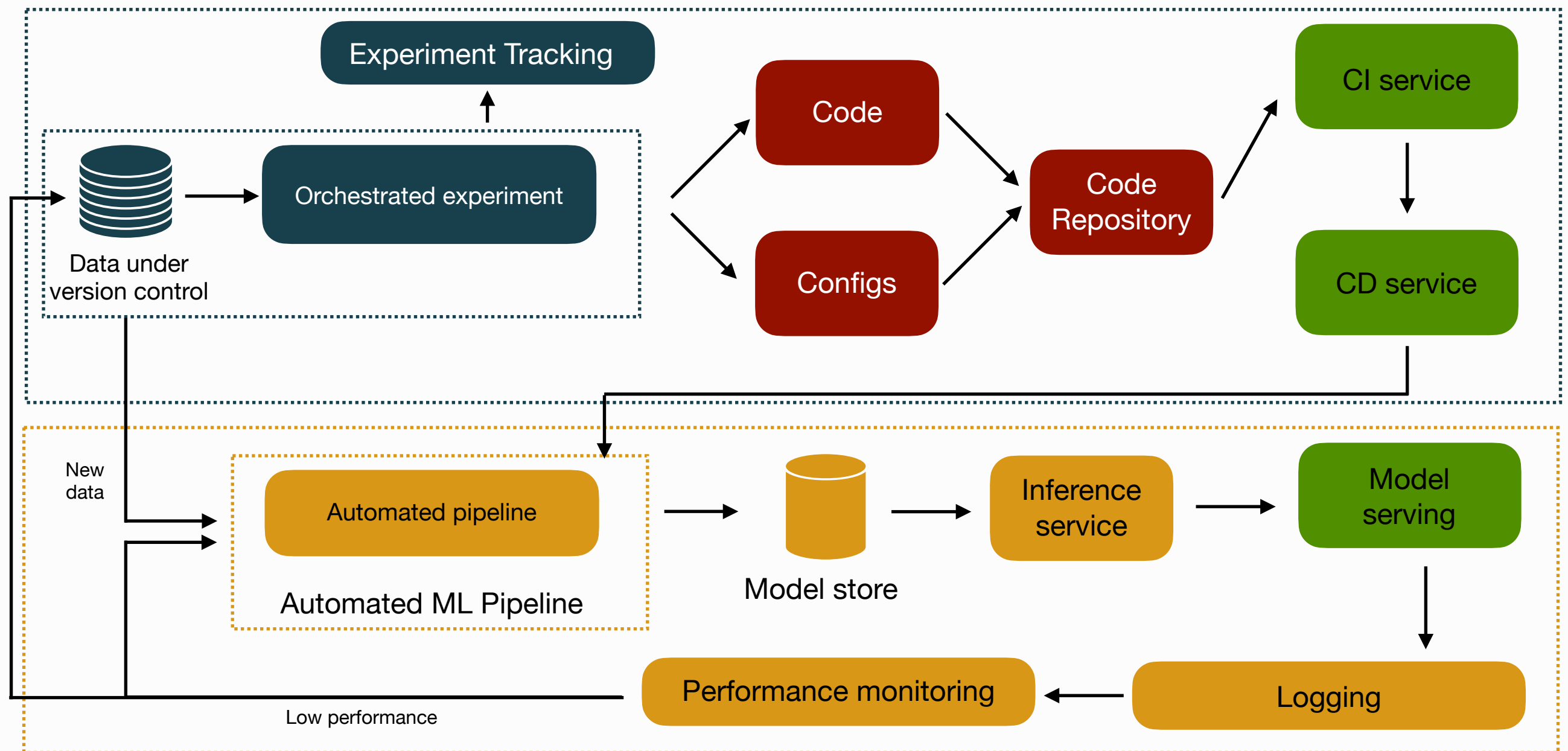


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With MLOps - Maturity Level 2

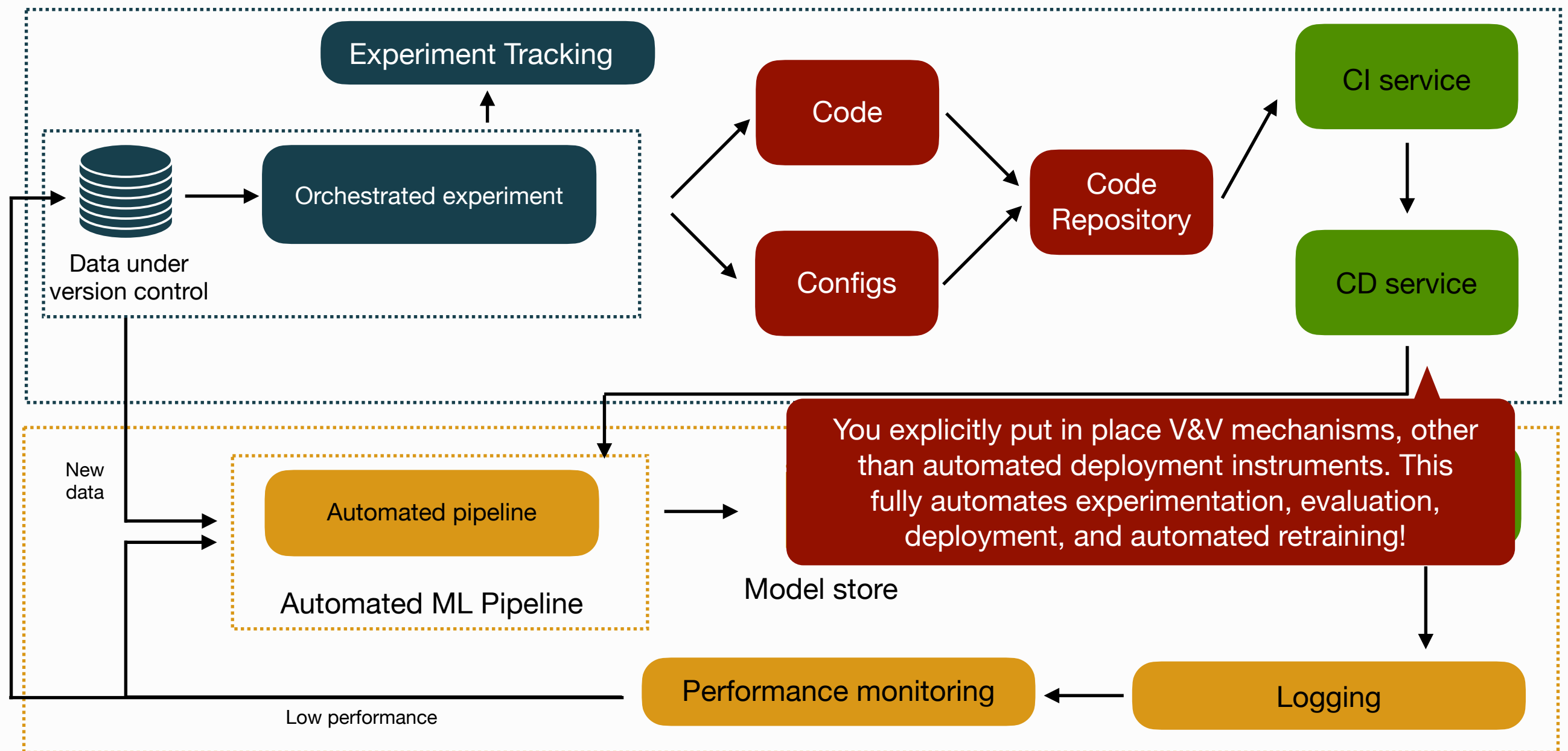


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With MLOps - Maturity Level 2

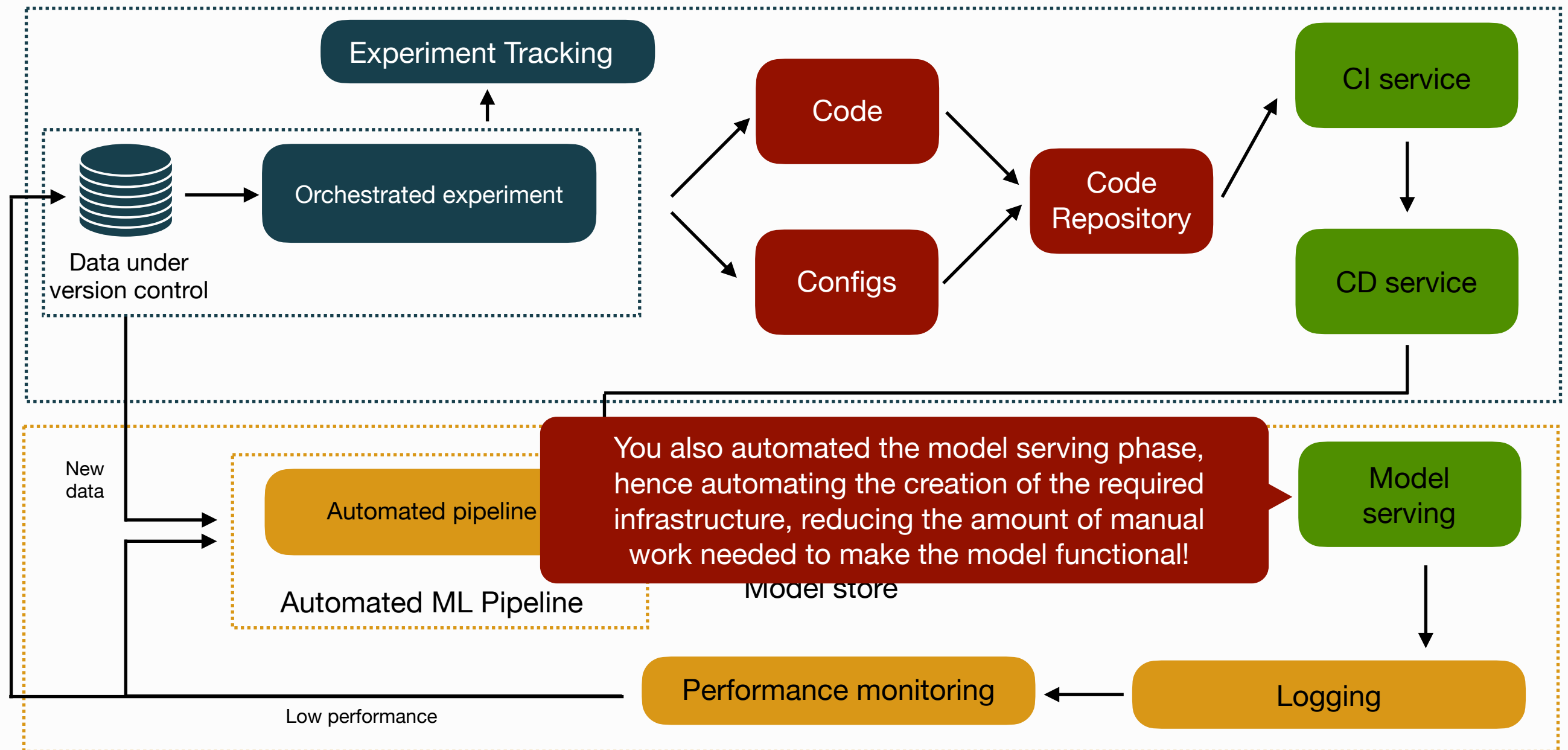


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With MLOps - Maturity Level 2

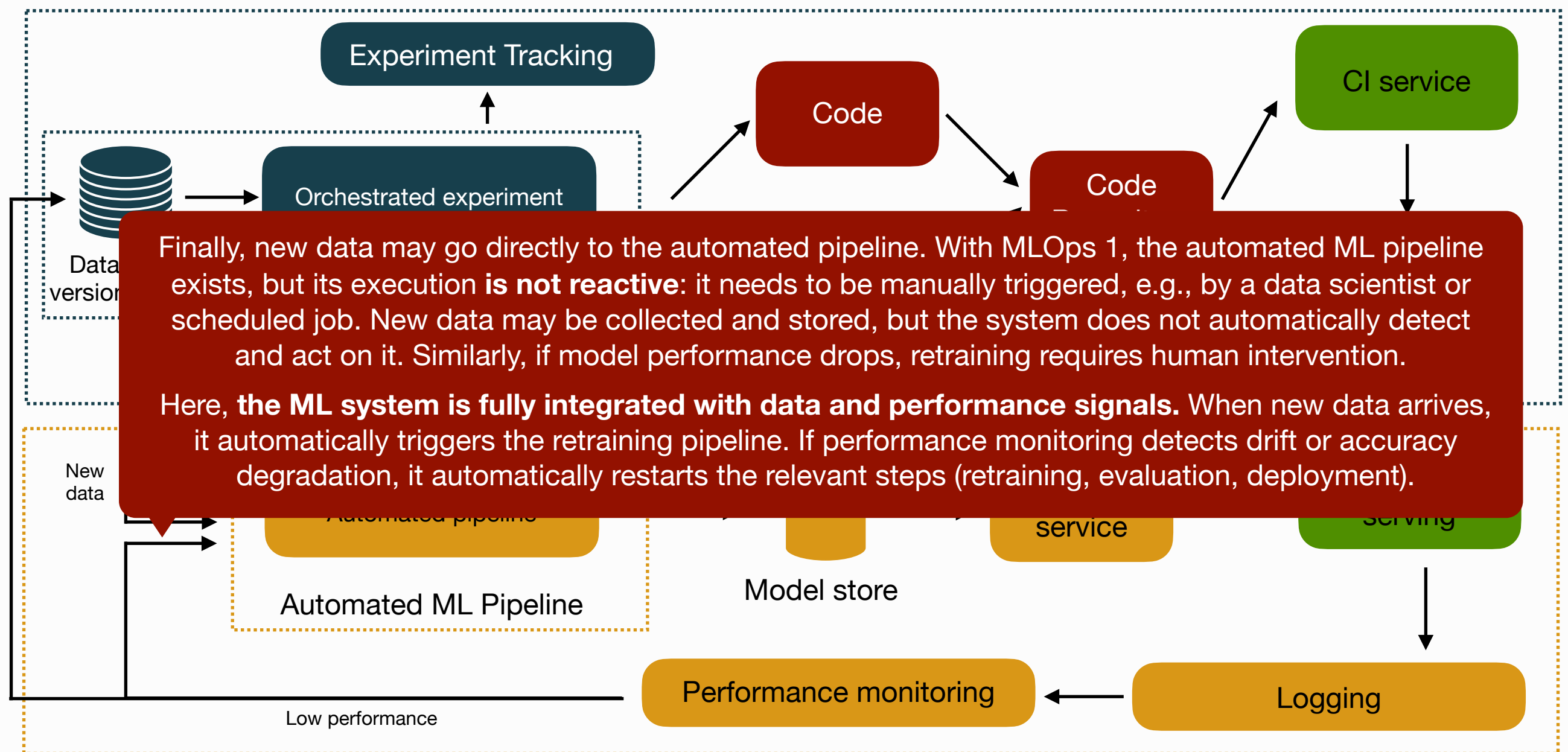


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With MLOps - Maturity Level 2



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To sum up:

- With MLOps at maturity level 0 you aim at making experiments and code reproducible, other than providing the trained model to machine learning engineers;
- With MLOps at maturity level 1 you aim at making you machine learning system able to react to performance drifts through automated retraining;
- With MLOps at maturity level 2 you aim at fully automating experimentation, validation & verification, and deployment of the model.

MLOps has natural implications for quality assurance: at maturity level 2, you can deploy automatic instruments to verify the presence of technical debt - much in the same way you would do with traditional systems. In addition, the entire system can automatically readjust to code, data, or performance changes.

In other terms, MLOps allows you to *treat a machine learning model as any other software component*, hence verifying its quality and validity over time.

As such, MLOps enables full control over the steps performed to preprocess data, train the model, and deploy it within a software engineering project.

Software Engineering for Artificial Intelligence: Introduction to MLOps

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

MLOps = From Observation to Action

AI in Operation was about seeing what's going on. MLOps is about acting on it automatically, efficiently, and reproducibly.

Data

AI Observability → Detection of Drift / Errors / Degradation

MLOps

MLOps Automations

Decisions

Retraining / Re-deployment / Model rollback / Alerting

Software Engineering for Artificial Intelligence: Introduction to MLOps

From AI-Enabled Systems Operation to MLOps

DevOps is a set of practices that combines software development and IT operations to enable faster, more reliable, and continuous delivery of software through automation, collaboration, and shared responsibility.

Two-week sprint begins:

- The data scientist retrains the model on recent data.
- The backend engineer wraps the model in a REST API.
- The frontend developer updates the homepage UI.
- The QA engineer defines tests for accuracy and performance.
- Daily stand-ups help coordinate progress and uncover blockers.

CI

CD

Continuous integration and delivery

More tools to improve responsiveness: essentially, a software-defined lifecycle extended to operation

Infrastructure code

Monitoring and Alerting

Incident Management and Feedback Loops

Software Engineering for Artificial Intelligence: Introduction to MLOps

From Model Lifecycle to MLOps

MLOps. MLOps refers to a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

Once again, it is important to remark that MLOps is not a single tool or platform, but rather it is about agreeing to machine learning the right way and then supporting it.

A few shared principles

Process

ML should be collaborative

ML should be reproducible

ML should be continuous

ML should be tested and monitored

Shared infrastructure

Versioning for code, data, and metadata

Machine learning pipelines

Model Deployment and Monitoring

Tooling

Software Engineering for Artificial Intelligence: Introduction to MLOps

From Model Lifecycle to MLOps

MLOps. MLOps refers to a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

With MLOps - Maturity Level 0

Experiment Tracking

Data under version control

Data validation and processing

Model training

Model evaluation and validation

Code

Configs

Code Repository

Experiment Orchestration

Model store

Inference service

Logging

Software Engineering for Artificial Intelligence: Introduction to MLOps

From Model Lifecycle to MLOps

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With MLOps - Maturity Level 1

Experiment Tracking

Data under version control

Data validation and processing

Model training

Model evaluation and validation

Code

Configs

Code Repository

Experiment Orchestration

Automated ML Pipeline

Model store

Inference service

Logging

Performance monitoring

Software Engineering for Artificial Intelligence: Introduction to MLOps

From Model Lifecycle to MLOps

MLOps. MLOps refers to a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

With MLOps - Maturity Level 2

Experiment Tracking

Data under version control

Orchestrated experiment

Code

Configs

Code Repository

CI service

CD service

Automated pipeline

Automated ML Pipeline

Model store

Inference service

Model serving

Logging

Performance monitoring



Software Engineering for Artificial Intelligence

Introduction to MLOps

